

US EPR — Plant Systems Control Reference

Source: US EPR FSAR (AREVA NP / NRC DCA), Chapters 5-11, 18. Compiled for custom MCR layout design.

MAIN — Main Control Room — Controlled via PICS workstations or SICS hardwired panel

System group	System name	Abbreviation	Description / function
Reactor core & reactivity control	Control rod drive control system	CRDCS	Rod cluster control assembly insertion/withdrawal, rod position indication via RPMS
	Reactor trip system - manual trip	RTS / SICS	Hardwired manual RT initiation from SICS; automatic trip from PS (TELEPERM XS)
	Neutron monitoring system	NMS / EIS	Source-range, intermediate-range, power-range ex-core neutron flux monitoring
	In-core instrumentation system	ICIS / SPND	72 self-powered neutron detectors for DNBR/LPD protection and flux mapping
	Reactor control, surveillance & limitation	RCSL	Automated rod control, axial offset alarm, turbine run-back, power limitations
	Anti-dilution mitigation system	ADM	Prevents inadvertent dilution of boron concentration at power
Reactor coolant system	Reactor coolant pumps	RCP (x4)	Start, stop, speed monitoring; seal injection flow; RCP trip on ECCS initiation
	Pressurizer level & pressure control	PZR	Heater group selection, spray valve control, continuous level indication
	Pressurizer power-operated relief valves	PORV / POSRV	Manual open/close; block valve position; LTOP monitoring
	RCS pressure boundary monitoring	RCS P/T	Loop Tc/Th, pressure, subcooling margin, loop flow - trend displays on PICS
Main steam system	Main steam isolation valves	MSIV (x4)	Open/close command from PICS and SICS; position indication; auto-closure on SI or MSLB
	Main steam safety/relief valves	MSSV / MSRT	Position indication; MSRT remote manual actuation
	Turbine bypass system / steam dump	TBS / TBV	Condenser dump valve control; SG pressure relief during load rejection or trip
	SG blowdown isolation & processing	SGBS	Blowdown isolation valve control; activity monitoring tie-in to PICS
Feedwater & steam generators	Main feedwater pumps (turbine-driven)	MFPT (x2)	Speed control, trip, auxiliary oil pump; SG level control via PICS/PAS
	Main feedwater flow & SG level	FW / SGL	Three-element level control; feedwater flow transmitters; isolation valve status
	Feedwater isolation valves	FWIV	Manual and auto close from PICS or SI actuation; position monitoring
	Feedwater heater extraction valves	FWH EXT	Extraction steam valves and drains managed via PICS turbine island displays
Emergency feedwater system	EFW motor-driven pump (x2)	EFWP-MD	Manual start/stop from MCR; SICS hardwired initiation; flow to all 4 SGs
	EFW turbine-driven pump	EFWP-TD	Remote manual steam admission; auto-start on low SG level / SI actuation
	EFW flow & isolation valves	EFWIV	Per-SG isolation valve open/close; flow indication; auto-recirculation

Chemical & volume control system	Charging pumps	CVCS-CP (x3)	Start/stop, flow control; normal charging, ECCS mode switching; RCP seal injection
	Boration & dilution control	CVCS B/D	Boric acid batching, dilution valve sequence; boron concentration setpoint
	Letdown system	CVCS-LD	Letdown orifice selection, holdup tank level, regenerative HX control
	Volume control tank	VCT	Level, pressure, N2 blanket; transfer pump control; chemistry tie-ins
	Boron concentration measurement	BCMS	Online boron concentration indication from grab-sample unit and inline measurement
Extra boration system	EBS initiation & control	EBS	Diverse boration path; manual actuation from SICS; automatically actuated on SAS signal
Safety injection / ECCS	SI actuation (manual ESF)	ESFAS / SICS	Manual initiation from SICS pushbuttons; automatic via PS; 4 divisions
	Medium-head safety injection pumps	MHSI (x4)	One per division; injection header valve status; automatic ECCS switchover
	Low-head safety injection pumps	LHSI (x4)	One per division; long-term recirculation; tie-in with RHR in recirculation mode
	Accumulators (passive injection)	ACC (x4)	N2 pressure and water level monitoring; isolation valve position indication
	IRWST level monitoring	IRWST	In-containment refuelling water storage tank inventory; gravity feed confirmation
Residual heat removal system	RHR pumps (x4, 1 per division)	RHRS-P	Start/stop; flow rate control; mode switching injection to recirculation
	RHR heat exchanger control	RHRS-HX	Bypass valve control for cooldown temperature rate; CCWS flow tie-in
	RHR suction & injection valve alignment	RHRS VLV	Manual or auto valve sequence for cold-leg / hot-leg recirculation alignment
Containment systems	Containment isolation actuation	CIA / SICS	Phase A/B isolation manual and automatic; position indication of all CIVs
	Containment spray system	CSS (x2)	Manual and auto actuation; spray pump start/stop; additive system monitoring
	Containment atmosphere monitoring	CAM	H2 concentration, temperature, pressure; post-accident display on PAMS/QDS
	Hydrogen monitoring & igniters	HMS / PAR	H2 sensor display; igniter enable/disable (SICS); passive autocatalytic recombiner status
	Containment vacuum relief valves	CVR	Monitoring of vacuum relief actuations; auto-reclose confirmation
Turbine-generator	Turbine startup, load & trip	TG / EHC	Electrohydraulic control via PICS; turbine run-up curve; automatic load control
	Generator excitation & voltage control	AVR	Automatic voltage regulator setpoint; field current monitoring; reactive power control
	Main generator breaker & tie	GEN CB	Breaker close/open command; synchronising check; grid power flow indication
Condenser & condensate	Main condenser / air extraction	COND / SGAE	Condenser pressure, hotwell level; steam jet air ejector / vacuum pump status
	Condensate pumps	CDP (x3)	Start/stop, flow, pressure; automatic recirculation valve control
	Condensate polishing system	CPS	Vessel-in-service selection; resin fouling alarms; conductivity / pH monitoring
Component cooling water system	CCWS pumps - status & alignment	CCWS-P	Pump run/standby status; auto-start on CCWS low-pressure signal; flow headers
	CCWS isolation valves (safety-related)	CCWS ISO	Manual and ESFAS-initiated isolation of loads; valve position display

Essential service water system	ESWS pumps - start/stop & alignment	ESWS-P (x4)	One per division; manual and auto start; flow indication; CCWS HX supply
	ESWS isolation & bypass valves	ESWS VLV	Division alignment valves; crosstie isolation; ultimate heat sink monitoring
AC/DC electrical power	Normal auxiliary transformers / switchyard	NAT / SWY	Grid voltage monitoring; NAT tap changer; breaker status; load-shedding actuation
	Class 1E AC switchgear (4 divisions)	EPSS 6.9kV	Bus tie breaker status; load-shedding sequence status; bus voltage/freq indication
	Emergency diesel generators	EDG (x4)	Auto-start confirmation; manual start/stop; output breaker; fuel oil level
	Class 1E UPS / battery (x4 per div)	EUPS / BAT	Battery voltage, charger status, inverter alarms; auto-transfer indication
	Non-Class 1E UPS	NUPS	Status and alarms for PICS/PAS/PACS power supplies
	Manual bus transfer (hardwired)	BUS XFR / SICS	SICS-hardwired manual bus transfer pushbuttons for each safeguard bus
HVAC	MCR emergency habitability - CRACS	CRACS	Mode selection (normal/recirc); positive pressure monitoring; charcoal filter bypass
	Nuclear island / safeguard building HVAC	NI HVAC	Supply/exhaust isolation valves; controlled-area negative pressure monitoring
	Fuel building HVAC isolation	FB HVAC	Isolation damper close on high radiation; filter train status
	Reactor building containment purge	RB HVAC	Containment purge isolation on high activity; atmosphere temperature monitoring
Radiation monitoring	Process & effluent radiation monitoring	PERMS / RMS	Main steam, condenser off-gas, gaseous waste, liquid waste, stack release rates
	Area radiation monitoring	ARM	Fixed area monitors throughout plant; high-rad alarms to MCR alarm workstation
	Airborne radioactivity monitoring	ABAM	Continuous air samplers in occupied areas; iodine/particulate indication
	Post-accident monitoring	PAMS / QDS	Qualified Display System - RCS subcooling, SG level, containment P/T, PZR level
	Seismic monitoring system	SMS	Free-field and structural accelerograph indication; SSE threshold alarm
Radioactive waste management	Liquid radwaste - tank & transfer	LRS	Tank level/activity indication; transfer pump status; sampling authorization
	Gaseous radwaste - storage & vent	GRS	Delay-bed tank pressure/activity; controlled release valve; stack monitor tie-in
	Solid radwaste monitoring	SRS	Drum/package inventory and radiation field alarms from radwaste building monitors
Compressed air & sampling	Instrument / service air pressure monitoring	IAS / SAS	Header pressure alarms; automatic switchover to backup N2 or backup compressor
	NSS & process sampling status	NSSS sampling	Sample isolation valve control; radioactive sample routing to MCR display
Spent fuel & fuel handling	SFP cooling system monitoring	SFPCS	Pool temperature, level; pump run status; makeup alignment via CCWS
Fire protection	Fire detection & alarm annunciation	FDS	Zone-by-zone fire alarm display; smoke/heat detector status on MCR annunciator
	Fire suppression actuation (key-op)	FSS	Key-operated MCR initiating stations for CO2/FM200 systems in critical rooms

AUX — Local / Auxiliary Control Stations — Operated at local panels or LCS outside the MCR

System group	System name	Abbreviation	Description / function
Reactor core & reactivity control	CRDM local wiring / connectors	CRDM local	Local conduit and connector boxes on reactor head; field access during outages only

Reactor coolant system	RCP motor control centres	<i>RCP MCC</i>	Local motor starters and overload protection panels in electrical equipment rooms
	Pressurizer heater local panels	<i>PZR HTR</i>	Local breaker panels for backup heater groups in the nuclear auxiliary building
Main steam system	MSIV local position indicators	<i>MSIV local</i>	Local position indicators on valve body in main steam tunnel for walkdowns
	Steam line drain valves	<i>MSDR local</i>	Warm-up drain valve manual handwheels in steam tunnel; locally operated during startup
Feedwater & steam generators	MFPT local control panel	<i>MFPT local</i>	Local lube-oil, turning-gear and trip-reset panel on feedwater pump deck
	Feedwater heater drain local controls	<i>FWH local</i>	Local handvalves and bypass connections on feedwater heater strings
Emergency feedwater system	EFW turbine local panel	<i>EFWP-TD local</i>	Local governor and hand-trip on pump turbine; backup hand-valve for steam supply
	EFW motor pump local MCCs	<i>EFWP-MD local</i>	Local disconnect and overload panels in safeguard building electrical rooms
Chemical & volume control system	CVCS local panel (charging pump area)	<i>CVCS local</i>	Local valve stations and sample connections near charging pumps in auxiliary building
	RCP seal system local	<i>RCP SEAL local</i>	Local pressure and temperature gauges on RCP seal injection/return lines
Extra borating system	EBS local panel	<i>EBS local</i>	Local pump/valve panel near IRWST/boron tanks in safeguard buildings
Safety injection / ECCS	SI pump local MCCs	<i>SI MCC</i>	Local motor control centres for MHSI/LHSI in safeguard building electrical rooms
	Accumulator local isolation valves	<i>ACC ISO local</i>	Local bypass and test valves on accumulator lines; maintenance isolation points
Residual heat removal system	RHR local valve panels	<i>RHRS local</i>	Local handstations on major RHR isolation and throttle valves in safeguard buildings
Containment systems	Containment spray pump local panels	<i>CSS local</i>	Local pump start/stop and discharge valve panels in safeguard pump buildings
	H2 recombiner local panels	<i>PAR local</i>	Local connection boxes for PAR modules distributed inside containment
	CIV local handstations	<i>CIV local</i>	Local (fail-safe) handstations on containment isolation valves for maintenance testing
Turbine-generator	Turbine local control panel	<i>TG local</i>	Local EHC workstation on turbine deck; manual turning gear, trip reset, lube oil
	Turbine lube oil system	<i>LOS local</i>	Local oil pump start/stop, bearing temp panels on turbine deck
	Generator H2 cooling & sealing	<i>GEN H2 local</i>	Local H2 pressure and purity panels beside generator stator
	Turning gear local	<i>TG turning gear</i>	Local engage/disengage and speed indication panel at turbine coupling end
Condenser & condensate	Circulating water system	<i>CWS local</i>	Local pump start/stop and sluice gate panels at intake structure and cooling tower
	Cooling tower fans & basin	<i>CT local</i>	Local fan cell start/stop, basin level, anti-freeze heater controls at cooling tower
	Condenser vacuum pump local	<i>COND local</i>	Local start/stop for vacuum priming pumps below turbine deck
	Condensate polisher local	<i>CPS local</i>	Local regeneration and resin-transfer control panel in condensate polisher building
Component cooling water system	CCWS local pump MCCs	<i>CCWS local</i>	Local motor control centres in safeguard/nuclear auxiliary buildings
	CCWS HX flow control local	<i>CCWS HX local</i>	Local temperature control valves and bypass handvalves at each CCWS HX

Essential service water system	ESWS pump local panels	<i>ESWS local</i>	Local pump houses at UHS/cooling pond; local start and flow gauges
	Trash rack / screen wash	<i>ESWS screen</i>	Local screen-wash motors and traveling screen controls at intake structure
AC/DC electrical power	MV/LV switchgear local panels	<i>SWGR local</i>	Local close/open and protection relay panels in electrical equipment rooms
	EDG local control panels	<i>EDG local</i>	Local start/stop, governor, voltage regulator panels in diesel generator buildings
	Battery charger local panels	<i>BAT local</i>	Local float/equalize selection and ammeter panels in battery rooms
	DC distribution local panels	<i>DC local</i>	Local 125 VDC / 24 VDC distribution boards in safeguard building electrical rooms
HVAC	AHU / fan local control panels	<i>AHU local</i>	Local start/stop and damper handstations on individual air handling units
	Chiller / DX unit local controls	<i>CHILLER local</i>	Local compressor and refrigerant panels for each chilled-water unit
	Filtered exhaust unit local	<i>FEU local</i>	Local fan start/stop and pressure differential panels on HEPA/charcoal filter trains
Radiation monitoring	Local radiation monitors (area-specific)	<i>RM local</i>	Wall-mounted analog/digital readout panels in radiation-controlled areas
Radioactive waste management	Liquid waste processing - local	<i>LWP local</i>	Local evaporator, ion-exchanger and filter panel in radwaste processing building
	Gas processing unit local	<i>GPU local</i>	Local compressor, charcoal delay bed and HEPA train controls in radwaste building
	Solid waste packaging local	<i>SWP local</i>	Local crane, press and drum filler controls in radwaste packaging area
Compressed air & sampling	Compressor local panels	<i>COMP local</i>	Local start/stop and load/unload controls at each air compressor
	Dryer & filter local	<i>CAS local</i>	Local dew-point indication and regeneration-cycle panels at air drying units
	Sample panel (chemistry lab area)	<i>SAMP local</i>	Local sample sink, grab-sample valves and flow meters near chemical laboratory
Spent fuel & fuel handling	SFP makeup & purification	<i>SFPCS local</i>	Local pump and valve panels for SFP cooling loop in fuel building
	Fuel handling machine	<i>FHM local</i>	Local operator console on fuel handling machine bridge in fuel/reactor building
	Cask decontamination & pit local	<i>CASK local</i>	Local cask pit crane controls and dose-rate meters in spent fuel cask loading area
Fire protection	Fire suppression local panels	<i>FSS local</i>	Local fire control panels in each building (nuclear island, turbine, radwaste, etc.)
	Diesel fire pump local	<i>DFP local</i>	Local auto-start and manual control panel at diesel-driven fire water pump house